

Functional Equation - Part 1

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Chapter 0

Preliminaries

0.1 Notations

- $\mathbb{N}$: the set of all positive integers
- $\mathbb{Z}$: the set of all integers
- $\mathbb{Q}$: the set of all rational numbers
- $\mathbb{R}$: the set of all real numbers
- $\mathbb{N}_0$: defined by $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$
- $\mathbb{R}^+$: the set of all positive reals

0.2 Character

Chapter 1

Basic Concepts

Solving functional equations is a main part in Olympiad Algebra. We will first introduce the basic ideas. For example, apply injectivity to reduce the equation. Or establish some inequalities to find the value of given point. Then, we're going to solve problems in past contests. Finally, we'll learn supplement materials, which are used to bash.

1.1 What are functional equations ?

We begin with a system of equations

$$\begin{aligned}2x - 5y &= 8 \\ 3x + 9y &= -12\end{aligned}$$

This is a special case of functional equations. To be precise, we translate the question to

Example 1. Find all functions $f : \{1, 2\} \rightarrow \mathbb{R}$ such that

$$\begin{aligned}2f(1) - 5f(2) &= 8 \\ 3f(1) + 9f(2) &= -12\end{aligned}$$

In fact, when the domain of f is a finite subset A of $\mathbb{N}$, it's just a system of equations with variables $\{f(a)\}_{a \in A}$. So, it's nature to consider finding all functions with particular properties as a generalization. Here is an example,

Example 2. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$2f(x) - 5f(y) = 8, \forall x, y \in \mathbb{R}$$

As dealing with systems of equations, we want to choose several suitable equations and solve. For instance, for $f(0)$, the simplest identity is

$$2f(0) - 5f(0) = 8$$

which gives $f(0) = -\frac{8}{3}$.

Example 3. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) + xf(1-x) = x$$

for all $x \in \mathbb{R}$

Solution. For any a , the identities in which $f(a)$ appears are

$$f(a) + af(1-a) = a$$

$$f(1-a) + (1-a)f(a) = 1-a$$

Direct to get $f(a) = \frac{a^2}{1-a+a^2}$. It's over? No, we should verify our answer. As following:

$$\begin{aligned} f(x) + xf(1-x) &= \frac{x^2}{1-x+x^2} + \frac{x(1-x)^2}{1-(1-x)+(1-x)^2} \\ &= \frac{x^2}{1-x+x^2} + \frac{x-2x^2+x^3}{1-x+x^2} \\ &= x \end{aligned}$$

Remark 1. The question, now, is why we need to check our answer. That's because, in most problems, we won't find all identities in which a given variable appears. The value may contradicts to other equations.

Remark 2. In fact, our central idea is:

- If $f(a)$ is unknown for some a , choose an equation in which $f(a)$ appears
- If the variables ($f(\text{something})$) in the chosen equation are known, go to the next step. Otherwise, back to Step 1.
- Solve the system of equations

1.2 Assertion

Example 4. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x - y) = f(x) + f(y) - 2xy$$

holds for all $x, y \in \mathbb{R}$

Solution. There are infinite equations being the concern of a given variable. We must choose some of them and solve it. Well, it's clear that for all $a \in \mathbb{R}$, we have

$$f(0) = 2f(a) - 2a^2$$

Once we know $f(0)$, we get $f(a)$. In fact, the above equation also tells us $f(0) = 0$ (why?) So the only possibility of the function is $f(a) = a^2$ for all a . Indeed,

$$f(x - y) = (x - y)^2 = x^2 + y^2 - 2xy = f(x) + f(y) - 2xy$$

Note that what we have done is to choose some equations and solve systems of equations. Equivalently, we assert values to variables. In the previous example, taking

$$f(0) = 2f(a) - 2a^2$$

and asserting $x = a, y = a$ are the same. As the result, I'll say "set $x = ?, y = ?$ " instead of "choose the following equation". Assertion is the easiest and most important step in solving functional equations. Consequently, let's do more examples. We'll see that for most problems, $f(0)$ is important.

Example 5. (Komal) Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y) + f(x)f(y) = x^2y^2 + 2xy \tag{1.1}$$

for all $x, y \in \mathbb{R}$

Solution. Set $y = 0$ into (1.1):

$$(1 + f(0))f(x) = 0$$

If $f(0) \neq -1$, then $f(x) = 0$ for all $x \in \mathbb{R}$, which is absurd. So $f(0)$ must be -1 .

Now, we set $(x, y) = (a, a), (a, -a), (2a, -a)$ to get

$$\begin{aligned} f(2a) + f(a)^2 &= a^4 + 2a^2 \\ f(a)f(-a) - 1 &= a^4 - 2a^2 \\ f(a) + f(2a)f(-a) &= 4a^4 - 4a^2 \end{aligned}$$

Those implies $f(a) = a^2 - 1$, which is indeed a answer to (1.1)

In this problem, we have solved the system of equations. In many "easy" functional equations, solving the system of equations is the key-point.

Example 6. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ so that

$$f(x+y) = f(x) + f(y) + 3(x+y) \sqrt[3]{f(x)f(y)}$$

holds for all $x, y \in \mathbb{N}$

Solution. Notice that

$$f(2x) = 2f(x) + \sqrt[3]{f(x)^2}$$

Therefore, we can find a function $g : \mathbb{N} \rightarrow \mathbb{N}$ defined by $g(x) = \sqrt[3]{f(x)}$ (why) Then g satisfies the following equation

$$g(x+y)^3 = g(x)^3 + g(y)^3 + 3(x+y)g(x)g(y)$$

for all $x, y \in \mathbb{N}$. If we know the values of $g(1)$ and $g(2)$, it's easy to show g is the identity map by mathematical induction. As the previous example suggested, when one encounters the relation between $h(x), h(y), h(xy), h(x+y)$ for some function h , it's quite useful to get $h(1), h(2)$ by taking the following assertions:

$$(x, y) = (1, 1), (1, 2), (1, 3), (2, 2)$$

The remains are left as an exercise.

Example 7. (2015 Baltic Way) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying for all reals x, y ,

$$|x|f(y) + yf(x) = f(xy) + f(x^2) + f(f(y))$$

Solution. First, consider the assertion $(x, y) = (1, 1)$, we get

$$2f(1) = 2f(1) + f(f(1)) \rightarrow f(f(1)) = 0$$

Then, take $x = 0, y = 1$:

$$f(0) = 2f(0) + f(f(1))$$

So $f(0) = 0$. It follows that for $a \geq 0$, $f(a) = 0$ by setting $(x, y) = (\sqrt{a}, 0)$

Finally, since $f(f(b)) = 0$ for all $b \in \mathbb{R}$ (why), $f(c) = -f(-1)c$ for all negatives c (why)

1.3 Injectivity and Surjectivity

Sometimes, it's not clear to show, after assertion, what f should be. Perhaps, there are too many f , we aren't sure that whether $f(a)$ appears in equations. Therefore, we hope to reduce the numbers of f . There are two ways: injectivity and surjectivity.

Definition 1. A function $f : \mathbb{A} \rightarrow \mathbb{B}$ is called injective if $f(a_1) = f(a_2)$ implies $a_1 = a_2$

Obviously, whenever we prove that a function is injective, the equation

$$f(\text{something } 1) = f(\text{something } 2)$$

becomes useful.

Example 8. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ that satisfy $f(0) = 1$ and

$$f(f(n)) = f(f(n+2) + 2) = n$$

for all integers n

Solution. Notice that $f(f(n)) = n$ implies f is injective (why) Therefore,

$$f(f(n)) = f(f(n+2) + 2) \rightarrow f(n) = f(n+2) + 2$$

Then, by mathematical induction, $f(2m) = 1 - 2m$ for all $m \in \mathbb{Z}$ Now, given an odd integer k , write $k = 2l + 1$. Our goal is to find the value of $f(2l + 1)$. However, note that

$$f(f(n)) = n \quad \text{and} \quad f(2m) = 1 - 2m$$

It follows that

$$f(2l + 1) = f(1 - 2(-l)) = f(f(2(-l))) = -2l = 1 - (2l + 1)$$

So we conclude that $f(n) = 1 - n$ for all $n \in \mathbb{Z}$

Remark 3. For convenience, let $P(x_1, x_2, \dots, x_n)$ be the assertion of a functional equation with n variables. Assume there are two distinct elements a_1, a_2 in the domain of the function f with $f(a_1) = f(a_2)$. We should derive a contradiction by assertions. For many cases, just compare two equations. For instance, $P(a_1, x_2, \dots, x_n)$ and $P(a_2, x_2, \dots, x_n)$.

In the previous remark, we see a standard way for proving injectivity: contradiction method. Although the following example can be solved by exchanging x, y , we still give a proof with injectivity.

Example 9. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$f(x + f(y)) = f(x) + y$$

for all $x, y \in \mathbb{Z}$

Solution. If $f(a) = f(b)$ for some $a, b \in \mathbb{Z}$. Compare $y = a, b$ in the equation:

$$f(x) + a = f(x + f(a)) = f(x + f(b)) = f(x) + b \rightarrow a = b$$

In other words, f is injective. As mentioned before, to use the injectivity, it suffices to choose an equation of the form

$$f(\text{something } 1) = f(\text{something } 2)$$

Indeed, one may assert $y = 0$ and get

$$f(x + f(0)) = f(x)$$

It follows that $f(0) = 0$. We then see $f(f(y)) = y$ by taking $x = 0$ in the equation. Plugging back to the equation,

$$f(x + y) = f(x) + f(y)$$

In particular, $f(x + 1) = f(x) + f(1)$, which implies

$$f(x) = xf(1)$$

Most of them do not satisfy the original equation, the only possibilities are

$$f(x) = x, \forall x \in \mathbb{Z} \quad \text{or} \quad f(x) = -x, \forall x \in \mathbb{Z}$$

Remark 4. Here is a useful result: If $g \circ f$ is injective, then so is f

Proof. Use the contradiction method, again. Suppose f is not injective, then there are two elements a, b such that $f(a) = f(b)$. So we have

$$g(f(a)) = g(f(b)) \rightarrow a = b$$

since $g \circ f$ is assumed to be injective. Thus, we conclude that f must be injective. □

For more information, https://en.wikipedia.org/wiki/Injective_function

Definition 2. A function $f : \mathbb{A} \rightarrow \mathbb{B}$ is called surjective if for any $b \in \mathbb{B}$, there exists $a \in \mathbb{A}$ satisfying $f(a) = b$. Equivalently, the pre-image of any element of $\mathbb{B}$ is not empty.

Example 10. Find all functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that g is injective and

$$f(g(x) + y) = g(x + f(y))$$

for all x, y in $\mathbb{R}$

Solution. Since g is injective, we tend to fix LHS. Thus, set $y = -g(x)$ to get

$$f(0) = g(x + f(-g(x)))$$

There must be a constant c such that

$$x + f(-g(x)) = c, \forall x \in \mathbb{R}$$

This equation, in particular, shows that f is surjective. Now, the original condition tells us g is also surjective. Let a be a real number with $g(a) = 0$. Then

$$f(y) = g(a + f(y)) \rightarrow y - a = g(y), \forall y \in \mathbb{R}$$

by using the surjectivity of f . Substitute into the original condition,

$$f(x + y - a) = x + f(y) - a$$

which is easy to get $f(x) = x, \forall x \in \mathbb{R}$ (left as an exercise)

Indeed, for every surjective function, we can replace f (something) to an arbitrary number whenever something can be any number of the codomain.

Definition 3. A function $f : \mathbb{A} \rightarrow \mathbb{B}$ is called bijective if f is both injective and surjective.

Sometimes, it's not obvious that a function is injective because the condition is complicated. However, if we know that the function is surjective, then we can reduce the condition to a simpler form. Then, maybe it's possible to show the injectivity.

Example 11. (2002 A1) Find all functions f from reals to reals such that

$$f(f(x) + y) = 2x + f(f(y) - x)$$

for all x, y in $\mathbb{R}$

Solution. Since there is already a term " $2x$ ", which is a surjective function. So if we want to show f is surjective, we may hope x is "free". Therefore, one should choose a suitable y such that one of the terms $f(f(x) + y)$ and $f(f(y) - x)$ is a constant.

$$P(x, -f(x)) \rightarrow f(f(-f(x)) - x) = f(0) - 2x$$

So f is surjective. Now, suppose that a, b are reals such that $f(a) = f(b)$. Then,

$$P(a, y), P(b, y) \rightarrow 2a + f(f(y) - a) = 2b + f(f(y) - b)$$

We hope $a = b$, which is equivalent to $f(f(y) - a) = f(f(y) - b)$ for some y . Note that, so far, we have only one known pair (x, y) such that $f(x) = f(y)$, namely (a, b)

Choose y to be the real with $f(y) = a + b$, the conclusion follows. Finally, one would like to set $x = 0$ (why)

$$f(f(0) + y) = f(f(y)) \rightarrow f(y) = y + f(0)$$

by using injectivity. Those functions are solutions as

$$\begin{aligned} LHS &= f(x + f(0) + y) = x + y + 2f(0) \\ &= 2x + f(y + f(0) - x) = 2x + f(f(y) - x) = RHS \end{aligned}$$

Example 12. (2015 Swiss) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for arbitrary $x, y \in \mathbb{R}$,

$$(y + 1)f(x) + f(xf(y) + f(x + y)) = y$$

Solution. First of all, rewrite the condition,

$$f(xf(y) + f(x + y)) = (1 - f(x))y - f(x)$$

So f is surjective (why) Now, assume $a, b \in \mathbb{R}$ s.t. $f(a) = f(b)$. Consider

$$P(0, a) \rightarrow (a + 1)f(0) + f(f(a)) = a$$

$$P(0, b) \rightarrow (b + 1)f(0) + f(f(b)) = b$$

If $f(0) \neq 1$, then f is injective. However, $f(0) \neq 1$ since

$$P(0, c) \rightarrow (c + 2)f(0) = c$$

where c is a real with $f(c) = 0$. We conclude that f is bijective. Next,

$$P(c, 0) \rightarrow f(cf(0)) = 0 = f(c)$$

It follows that $f(0) = 0$. From here, one can easily find the desired function f is

$$f(x) = -x, \quad \forall x \in \mathbb{R}$$

1.4 Cauchy's Functional Equation

So far, one may observe that the domains of functional equations are usually " $\mathbb{R}$ ". Thus, I would like to introduce a result of real functions.

Definition 4. A function $f : \mathbb{A} \rightarrow \mathbb{B}$ is called additive if

$$f(a_1 + a_2) = f(a_1) + f(a_2)$$

holds for all $a_1, a_2, a_1 + a_2 \in \mathbb{A}$

We consider the situation that $\mathbb{A} = \mathbb{B} = \mathbb{R}$. By using induction, it's easy to check

$$f(n) = nf(1)$$

for all $n \in \mathbb{Z}$. (ref. example 9) For rationals, we have

$$f\left(\frac{p}{q}\right) = \frac{1}{q}f(p) = \frac{p}{q}f(1)$$

The first equality follows from $f(nx) = nf(x)$, which can be proved by induction, also. Now, we may expect that $f(x) = xf(1)$ for all $x \in \mathbb{R}$. However, it's false! (why?) Fortunately, we have a criterion.

Theorem 1. Any one of following conditions implies an additive $f : \mathbb{R} \rightarrow \mathbb{R}$ must be linear.

- f is continuous on some interval
- f is bounded on some interval
- f is monotonic on some interval

Proof. For the first one, notice that if f is continuous at one point, then f is continuous everywhere. Next, for any real r , choose a sequence of rationals $\{a_n\}_{n \geq 1}$ converging to that number. Since f is continuous,

$$f(rx) = \lim_{n \rightarrow \infty} f(a_n x) = \lim_{n \rightarrow \infty} a_n f(x) = r f(x)$$

For the second one, suppose there is a real r s.t. $f(r) \neq rf(1)$. WLOG f is increasing and $f(r) > rf(1)$. Choose a rational q so that $\frac{f(r)}{f(1)} > q > r$ then

$$f(r) < f(q) = qf(1) < f(r)$$

a contradiction. For the third, let a_n to be any rationals such that $nx - a_n \in [a, b]$. Then

$$|f(nx - a_n)| = |n(f(x) - xf(1)) - (a_n - nx)f(1)| \geq n|f(x) - xf(1)| - |(nx - a_n)f(1)|$$

which means $f(x)$ must be $xf(1)$, done! □

Note that the method that we used to prove the second (or third) part of theorem 1 is useful. That's because it's usually easier to determine the function on rationals and one can use rationals to approach any real number.

Example 13. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all reals x, y , we have

$$f(x)^2 + f(y)^2 + f((x+y)^2) = 2x^2 + 2y^2 + f(xf(y) + yf(x))$$

Solution. The original equation is complicate, especially the term $f(xf(y) + yf(x))$. To kick it out, we may hope to have $f(0) = 0$ (why) This is indeed the case,

$$P(0,0) \rightarrow 2f(0)^2 + f(0) = f(0)$$

implies $f(0) = 0$. Next, subtract $P(x,0)$ and $P(y,0)$ from $P(x,y)$, we have the following

$$Q(x,y) : f((x+y)^2) = f(x^2) + f(y^2) + f(xf(y) + yf(x))$$

Then, compare

$$P(x,0) : f(x)^2 + f(x^2) = 2x^2$$

$$P(-x,0) : f(-x)^2 + f(x^2) = 2x^2$$

we see that $f(-x) = \pm f(x)$ holds for all x . Assume $f(a) = f(-a) = 0$ for some a

$$P(a,-a) \rightarrow 0 = 4a^2 \Leftrightarrow a = 0$$

Now, if $f(b) = f(-b)$ for some b

$$Q(b,-b) \rightarrow 0 = 2f(b^2)$$

so $b^2 = 0$. In other words, $b = 0$. From here, one can conclude that f must be odd. Finally, add two equations $Q(x,y)$ and $Q(x,-y)$,

$$f((x+y)^2) + f((x-y)^2) = 2f(x^2) + 2f(y^2)$$

If we set $g(x) = f(x^2)$, it becomes

$$g(x+y) + g(x-y) = 2g(x) + 2g(y)$$

The remains are left to readers:

Exercise. 1. Show that $g(nx) = n^2g(x)$ for all $n \in \mathbb{N}$ and $x \in \mathbb{R}$

2. Notice that $0 \leq g(x) \leq 2x^2$. Try to prove that $g(x) = g(1)x^2$ holds for all $x \in \mathbb{R}$

We return to an application of the theorem.

Example 14. (2015 Korea) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^{2015} + f(y)^{2015}) = f(x)^{2015} + y^{2015}$$

holds for all reals x, y

Solution. Start with

$$f(f(f(x^{2015} + f(y)^{2015}))) = f(f(f(x)^{2015} + y^{2015})) = x^{2015} + f(y)^{2015}$$

By using the surjectivity of x^{2015} , we get $f(f(x)) = x$ and thus, f is bijective. Now,

$$P(x, f(0)) \rightarrow f(x^{2015}) = f(x)^{2015} + f(0)^{2015}$$

and then

$$P(x, f(y)) \rightarrow f(x^{2015} + y^{2015}) = f(x)^{2015} + f(y)^{2015} = f(x^{2015}) + f(y^{2015}) - 2f(0)^{2015}$$

Again, use the surjectivity of x^{2015} , the above becomes

$$f(x + y) = f(x) + f(y) - 2f(0)^{2015}$$

One should be able to determine the value of a (hope so) Thus, we have

$$f(x + y) = f(x) + f(y) \quad \text{and} \quad f(x^{2015}) = f(x)^{2015}$$

We claim that f is linear: Given any real number x . We have, for any rational number q ,

$$f(x + q)^{2015} = (f(x) + f(q))^{2015} = f(x)^{2015} q f(1) f(x)^{2014} + \dots + q^{2015} f(1)^{2015}$$

On the other hand,

$$f(x + q)^{2015} = f((x + q)^{2015}) = f(x)^{2015} + 2015 q f(x)^{2014} + \dots + q^{2015} f(1)^{2015}$$

Since q can be arbitrary, $f(x^{2014}) = f(1) f(x)^{2014}$ (why) In particular, the second condition in the theorem is fulfilled. The conclusion follows.

Example 15. (2017 CSP) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that

$$f(x) - f(x+y) = f\left(\frac{x}{y}\right) f(x+y)$$

holds for all $x, y \in \mathbb{R}^+$

Solution. Because we expect the only non-trivial solution is

$$f(x) = \frac{1}{x}, \quad \forall x \in \mathbb{R}^+$$

we want to show that $f(x) \neq 0$ for all $x \in \mathbb{R}^+$

Assume $a \in \mathbb{R}^+$ so that $f(a) = 0$. Given $b < a$,

$$P(b, a-b) \rightarrow f(b) = 0$$

Also, notice that

$$P(a, \text{large}) \rightarrow f(\text{large}) = 0$$

Combine those results, we conclude that $f \equiv 0$

Now, suppose $f(x) \neq 0$ for all $x \in \mathbb{R}^+$. It's easy to see $f(1) = 1$ (why) So

$$P(x, x) \rightarrow f(x) = 2f(2x)$$

We have

$$P(x, y), P(y, x) \rightarrow \frac{f(x)}{f(y)} = \frac{f\left(\frac{x}{y}\right) + 1}{f\left(\frac{y}{x}\right) + 1}$$

and thus,

$$\frac{f\left(\frac{x}{y}\right) + 1}{2f\left(\frac{y}{x}\right) + 1} = \frac{f(x)}{2f(y)} = \frac{f(2x)}{f(y)} = \frac{\frac{1}{2}f\left(\frac{x}{y}\right) + 1}{2f\left(\frac{y}{x}\right) + 1} \Leftrightarrow f\left(\frac{x}{y}\right) f\left(\frac{y}{x}\right) = 1$$

It follows that f is multiplicative and

$$f(x) f(y) = (f(x) + f(y)) f(x+y)$$

By using theorem 1, we know that (why)

$$f(x) = \frac{1}{x}, \quad \forall x \in \mathbb{R}^+$$

1.5 Establishing Inequalities

Sometimes, it's not straightforward to find a proof of $f(a) = b$ for some given a, b . Instead, we try to prove the following inequalities:

$$f(a) \geq b \quad \text{and} \quad f(a) \leq b$$

which may be much easier to handle.

Example 16. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ with $f(1) = 1$ and

$$f(x+5) \geq f(x) + 5 \quad \text{and} \quad f(x+1) \leq f(x) + 1$$

holds for all $x \in \mathbb{N}$

Solution. By the second inequality,

$$f(x+5) \leq f(x) + 5$$

Combine it with the first inequality, we get

$$f(x) + 5 \leq f(x+5) \leq f(x) + 5$$

So the equality of $f(x+5) \leq f(x) + 5$ must hold, which implies $f(x+1) = f(x) + 1$. A simple induction will give $f(n) = n$ for all $n \in \mathbb{N}$

Because the condition is already an inequality, it's very straightforward to come up with such a solution.

Exercise. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) + f(y) \geq 2f\left(\frac{x+y}{2}\right)$$

holds for all $x, y \in \mathbb{R}$

Let's, now, do a "true" functional equation.

Example 17. (2017 OAO Shortlist) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$\sum_{i=1}^n \frac{1}{f^i(n)} = 1$$

holds for all natural number n .

Solution. We want to find two relative equations. Here is a simple way, compare $P(n), P(f(n))$

$$\sum_{i=1}^n \frac{1}{f^i(n)} = 1 = \sum_{i=1}^{f(n)} \frac{1}{f^i(f(n))}$$

From above, we get that $f(n) > n - 1$ (why). In other words, $f(n) \geq n$ for all $n \in \mathbb{N}$. Therefore,

$$1 = \sum_{i=1}^n \frac{1}{f^i(n)} \leq \sum_{i=1}^n \frac{1}{n} = 1$$

It follows that $f(n) = n$ must hold for all natural n , which is clearly a solution of the functional equation.

Exercise. (Socrates) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for each positive integer n , there exists a positive integer k with

$$\sum_{i=1}^k f^i(n) = kn$$

To establish an inequality, we should take large numbers into consideration. For instance,

Example 18. (2014 Taiwan) Find all increasing functions $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfying $f(2) = 7$,

$$f(mn) = f(m) + f(n) + f(m)f(n)$$

for all positive integers m and n

Solution. Recall the well-known identity:

$$(x+1)(y+1) = xy + x + y + 1$$

One tends to rewrite the condition:

$$f(mn) + 1 = (f(m) + 1)(f(n) + 1)$$

Let $g(n) := f(n) + 1$, we then have $g(mn) = g(m)g(n)$ holds for all $m, n \in \mathbb{N}$. Now,

$$n^x > 2^y \Leftrightarrow \log_2 n > \frac{y}{x} \Leftrightarrow g(n) = g(n^x)^{\frac{1}{x}} > g(2^y)^{\frac{1}{x}} = 2^{\frac{3y}{x}}$$

This gives the lower-bound of $g(n)$, namely n^3 . A similar argument gives $g(n) \leq n^3$

Note that whenever the codomain of the function is $\mathbb{R}^+$ or $\mathbb{N}$, we have a trivial inequality.

Example 19. (2016 Poland) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$2f(f(f(x))) + 5f(f(x)) = f(x) + 6x$$

holds for all $x \in \mathbb{R}^+$

Solution. If one does not familiar to recurrence, take a look: https://en.wikipedia.org/wiki/Recurrence_relation. Given any $x \in \mathbb{R}$. Let $a_i = f^i(x)$. Then conditions $P(x), P(f(x)), \dots$ becomes

$$2a_{i+3} + 5a_{i+2} = a_{i+1} + 6a_i$$

for all $i \in \mathbb{N}_0$. To get the general formula of a_i , we should solve

$$2x^3 + 5x^2 - x - 6 = 0$$

The roots are $1, -2, -\frac{3}{2}$, $a_i = b + c(-2)^i + d(-\frac{3}{2})^i$ If b or c is not zero, then $a_i < 0$ for some integer i (why) This means the sequence $\{a_i\}_{i \geq 0}$ is constant. In other words, $f(x) = x$. Note that the choice of x can be arbitrary, we conclude the answer to this problem is

$$f(x) = x, \forall x \in \mathbb{R}^+$$

In most problems, one needs to establish the recurrence.

Example 20. (Socrates) Determine all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(f(x) + y) + f(x + y) = 2x + 2f(y)$$

holds for all $x, y \in \mathbb{R}^+$

Solution. We have the following equations:

$$P(x, f(1)) \rightarrow f(f(x) + f(1)) + f(x + f(1)) = 2x + 2f(f(1))$$

$$P(x, 1) \rightarrow f(f(x) + 1) + f(x + 1) = 2x + 2f(1)$$

$$P(1, x) \rightarrow f(x + f(1)) + f(1 + x) = 2 + 2f(x)$$

$$P(1, f(x)) \rightarrow f(f(x) + f(1)) + f(1 + f(x)) = 2 + 2f(f(x))$$

So,

$$f(f(x)) = 2x - f(x) + (f(f(1)) + f(1) - 2)$$

Denote $f(f(1)) + f(1) - 2$ by c , then

$$f(f(x)) = 2x - f(x) + c$$

A simple induction gives

$$f^n(x) = \frac{1}{3} \left(f(x) + 2x + nc - \frac{c}{3} \right) + \frac{1}{3} (-2)^n \left(x - f(x) + \frac{c}{3} \right)$$

The remains are easy.

1.6 Exercise

To get a better feeling, I strongly suggest that the readers do lots of functional equations because we usually come up with a great idea due to our experience.

1.6.1 2014

Problem 1. (*Baltic Way*) Find all functions f defined on all reals and taking real values such that

$$f(f(y)) + f(x - y) = f(xf(y) - x)$$

for all real numbers x, y

Problem 2. (*ELMO*) Find all triples (f, g, h) of injective functions from the set of real numbers to itself satisfying

$$f(x + f(y)) = g(x) + h(y)$$

$$g(x + g(y)) = h(x) + f(y)$$

$$h(x + h(y)) = f(x) + g(y)$$

for all real numbers x and y

Problem 3. (*IZHO*) Does there exist a surjective function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(f(x)) = (x - 1)f(x) + 2$$

for all real x

Problem 4. (*Kazakhstan*) Find all functions $f : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ such that for all $x, y, z \in \mathbb{Q}$,

$$f(x, y) + f(y, z) + f(z, x) = f(0, x + y + z)$$

Problem 5. (*USA*) Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$xf(2f(y) - x) + y^2f(2x - f(y)) = \frac{f(x)^2}{x} + f(yf(y))$$

holds for all $x, y \in \mathbb{Z}$ with $x \neq 0$

1.6.2 2015

Problem 6. (Belarus) Find all functions $f(x)$ determined on interval $[0, 1]$ satisfying

$$f(f(x)) = f(x)$$

$$\{f(x)\} \sin^2 x + \{x\} \cos f(x) \cos x = f(x)$$

Problem 7. (Canada) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$(n-1)^2 < f(n) f(f(n)) < n^2 + n$$

for every positive integer n

Problem 8. (Greece) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$yf(x) + f(y) \geq f(xy)$$

for all $x, y \in \mathbb{R}$

Problem 9. (Turkey) Find all the functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2) + 4xy^2 f(y) = (f(x-y) + y^2)(f(x+y) + f(y))$$

holds for every $x, y \in \mathbb{R}$

1.6.3 2016

Problem 10. (Balkan) Find all injective functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for every real number x and every positive integer n ,

$$\left| \sum_{i=1}^n i(f(x+i+1) - f(f(x+i))) \right| < 2016$$

Problem 11. (EMMO) Find all functions $f : \mathbb{N} \rightarrow \mathbb{R}$ such that

$$\sum_{k=0}^{n-1} \left\lfloor x + \frac{f(k)}{n} \right\rfloor = \lfloor xf(n) \rfloor$$

holds for all $x \in \mathbb{R}$ and $n \in \mathbb{N}$

Problem 12. (HMMT) Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function such that for $w, x, y, z \in \mathbb{N}$

$$f(f(f(z))) f(wxf(yf(z))) = z^2 f(xf(y)) f(w)$$

Show that $f(n!) \geq n!$ for every positive integer n

Problem 13. (Netherland) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy - 1) + f(x)f(y) = 2xy - 1$$

holds for all $x, y \in \mathbb{R}$

Problem 14. (Norway) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x)f(y) = |x - y| \cdot f\left(\frac{xy + 1}{x - y}\right)$$

holds for all $x \neq y \in \mathbb{R}$

Problem 15. (Romania) Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(a^2) - f(b^2) \leq (f(a) + b)(a - f(b))$$

holds for all $x, y \in \mathbb{R}$

1.6.4 2017

Problem 16. (Belarus) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ satisfying the following equation

$$f(x + f(xy)) = xf(1 + f(y))$$

Problem 17. (Greece) Find all functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(1) = -8$ and

$$f(x - 3f(y)) = xf(y) - yf(x) + g(x)$$

holds for all $x, y \in \mathbb{R}$

Problem 18. (IZHO) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$(x + y^2)f(yf(x)) = xyf(y^2 + f(x))$$

holds for all $x, y \in \mathbb{R}$

Problem 19. (Pakistan) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that for all distinct x, y, z ,

$$f(x^2) - f(y)f(z) = f(x^y)f(y)f(z)(f(y^z) - f(z^x))$$

Problem 20. (Turkey) Given a real number a , try to find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ so that

$$f(xy + f(y)) = yf(x) + a$$

for every $x, y \in \mathbb{R}$

Chapter 2

Cauchy FE

In chapter one, we have introduced some terminologies. Recall the statement of theorem 1: For an additive $f : \mathbb{R} \rightarrow \mathbb{R}$, any of the following conditions guarantees that f is linear.

- f is continuous on some interval
- f is bounded on some interval
- f is monotonic on some interval

We're not contented because additivity and others are strong conditions.

2.1 Taiwanese Transformation

If one makes a suitable transformation, then he/she will get another useful result.

Example 21. All monotonic functions $f : \mathbb{R} \rightarrow \mathbb{R}^+$ such that

$$f(x+y) = f(x)f(y)$$

are $f(x) = a^x, \forall x \in \mathbb{R}$ for some constant a .

Solution. Define $g : \mathbb{R} \rightarrow \mathbb{R}_{>a}$ by $g(x) = \log f(x)$. It's also monotonic since it's the composition of two monotonic functions. Now, the original equation becomes

$$g(x+y) = g(x) + g(y)$$

We deduce $g(x) = cx, \forall x \in \mathbb{R}$ for some constant c , which means

$$f(x) = a^x, \forall x \in \mathbb{R}$$

for some constant a (why)

Exercise. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ so that

$$f(xy) = f(x)f(y)$$

for all $x, y \in \mathbb{R}$. PS: Such a function is called multiplicative.

Remark 5. A multiplicative function satisfying Cauchy condition must be zero or identity.

Proof. Notice that if a function is multiplicative, then

$$f(x^2) = f(x)^2 \geq 0$$

So f is bounded on some interval. It follows that f is linear. □

Exercise. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ so that

$$f(xy) = f(x) + f(y)$$

for all $x, y \in \mathbb{R}$

Example 22. Given a constant a . Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ so that

$$f(x+y) = f(x) + f(y) + a$$

holds for all $x, y \in \mathbb{R}^+$

Solution. Why we want to transform the function? Because we hope to show that f is linear by using **Theorem 1**! Thus, after guessing the answer to the FE, we should apply the corresponded transformation. In this example, one may guess

$$f(x) = cx - a, \quad \forall x \in \mathbb{R}$$

for some constant $c \in \mathbb{R}$ are the only functions satisfying the conditions. So, we tend to define $g : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that

$$g(x) = f(x) + a$$

Then the identity becomes

$$g(x+y) = g(x) + g(y)$$

which is trivial?

Another example,

Example 23. Given a constant d . Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ so that

$$f(x + y + d) = f(x) + f(y)$$

holds for $x, y \in \mathbb{R}$

Solution. This is similar to the previous one. First of all, one need to guess the answer. Assume that f is linear:

$$f(x) = ax + b$$

for some constant $a, b \in \mathbb{R}$. Then we must have

$$ax + ay + ad + b = LHS = RHS = ax + ay + 2b \Leftrightarrow b = ad$$

In other words,

$$f(x) = cx + cd = c(x + d), \forall x \in \mathbb{R}$$

for some constant $c \in \mathbb{R}$. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by the following rule

$$g(x) = f(x + d) \rightarrow g(x + y) = g(x) + g(y)$$

Exercise. Given two real numbers a, b . Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y + a) = f(x) + f(y) + b$$

holds for all pairs $(x, y) \in \mathbb{R} \times \mathbb{R}$

Before the next subsection, I would like to summary our previous work. (under a suitable situation i.e., bounded)

Condition	Transformation	Solution
$f(x + y) = f(x) + f(y)$		cx
$f(xy) = f(x)f(y)$	$g(x) = \log f(x)$	a^x
$f(xy) = f(x)f(y)$	$g(x) = \log f(a^x)$	$x^{\log_a b}$
$f(xy) = f(x) + f(y)$	$g(x) = f(a^x)$	$\log_a x$
$f(x + y) = f(x) + f(y) + a$	$g(x) = f(x) + a$	$cx - a$
$f(x + y + a) = f(x) + f(y)$	$g(x) = f(x + a)$	$c(x + a)$
$f(x + y + a) = f(x) + f(y) + b$	$g(x) = f(x + a) + b$	$c(x + a) - b$

2.2 Variation

Return to additive functions, we have proved that under suitable conditions, they must be linear. In this section, we'll investigate more "suitable conditions".

Example 24. Find all additive functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f\left(\frac{1}{x}\right) f(x) = 1, \forall x \in \mathbb{R}^*$$

Solution. Notice that if f is a solution, then $-f$ is also satisfies the conditions. So, let's assume that $f(1) = 1$. We'll calculate $f\left(\frac{1}{x^2+x}\right)$ in two different ways.

$$f\left(\frac{1}{x^2+x}\right) = f\left(\frac{1}{x}\right) - f\left(\frac{1}{x+1}\right) = \frac{1}{f(x)} - \frac{1}{f(x)+1} = \frac{1}{f(x)^2 + f(x)}$$

On the other hand,

$$f\left(\frac{1}{x^2+x}\right) = \frac{1}{f(x^2+x)} = \frac{1}{f(x^2) + f(x)}$$

It follows that

$$f(x^2) = f(x)^2, \forall x \in \mathbb{R}$$

And therefore, by the corollary, f must be the identity.

This calculation is tricky, but it helps in many variations. More examples,

Example 25. (2008 Korea) Find all additive functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f\left(\frac{1}{x}\right) = \frac{f(x)}{x^2}, \forall x \in \mathbb{R}^*$$

Solution. This one looks very similar to the previous. We still assume $f(1) = 1$ and calculate the same expression $f\left(\frac{1}{x^2+x}\right)$.

$$f\left(\frac{1}{x^2+x}\right) = f\left(\frac{1}{x}\right) - f\left(\frac{1}{x+1}\right) = \frac{f(x)}{x^2} - \frac{f(x)+1}{(x+1)^2} = \frac{(2x+1)f(x) - x^2}{(x^2+x)^2}$$

It equals to

$$\frac{f(x^2+x)}{(x^2+x)^2} = \frac{f(x^2) + f(x)}{(x^2+x)^2}$$

which is equivalent to

$$f(x^2) + x^2 = 2xf(x)$$

Now, it's not obvious to show f is the identity. So we tend to calculate another expression

$$f\left(\left(x + \frac{1}{x}\right)^2\right) = f\left(x^2 + 2 + \frac{1}{x^2}\right) = 2xf(x) - x^2 + 2 + \frac{2f\left(\frac{1}{x}\right)}{x} - \frac{1}{x^2}$$

From early result, it must equal to

$$2\left(x + \frac{1}{x}\right)f\left(x + \frac{1}{x}\right) - x^2 - 2 - \frac{1}{x^2}$$

After reduction, we find that $f(x) = x, \forall x \in \mathbb{R}$

Example 26. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+1) = f(x) + 1, f\left(\frac{1}{x}\right)f(x) = 1, \forall x \in \mathbb{R}^*$$

2.3 Extension

Sometimes, we have a **partial** Cauchy's condition.

Example 27. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) = f(x) + f(y), \forall x \in \mathbb{R} \text{ and } y \in \mathbb{R}^+$$

Solution. For any $a, b \in \mathbb{R}$, choose a large $c > 0$ with $a+b+2c, b+2c > 0$, we have

$$f(a+b+2c) = f(a+b) + f(2c)$$

On the other hand,

$$f(a+b+2c) = f(a) + f(b+2c) = f(a) + f(b) + f(2c)$$

Thus, f is additive and must be linear.

To derive the Cauchy's condition from a partial Cauchy's condition, we usually take another sufficiently nice number and count a specify expression in two different ways.

Example 28. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ so that

$$f(x+y) = f(x) + f(y)$$

for all $x, y \in \mathbb{R}$ satisfying $|x-y| \leq 1$.

Solution. I'll show that $f(x+y) = f(x) + f(y)$ for all $|x-y| \leq 2$. Indeed,

$$2f\left(\frac{x+z}{2}\right) = f(x) + f(z), 2f\left(\frac{y+z}{2}\right) = f(y) + f(z)$$

where $z = \frac{x+y}{2}$. Now, notice that

$$2f\left(\frac{x+z}{2}\right) + 2f\left(\frac{y+z}{2}\right) = 2f\left(\frac{x+y+2z}{2}\right) = 2f(2z) = 4f(z)$$

Therefore,

$$f(x) + f(y) = 4f(z) - 2f(z) = 2f(z) = f(x+y)$$

One can use induction to establish the Cauchy's condition.

As the previous example suggests, extending several times may be much easier

Example 29. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ satisfying the equation

$$f(z+x) + f(z+y) = f(2z+x+y), \forall x, y, z \in \mathbb{R}^+ \text{ with } x, y < z^2$$

Solution. I'll use induction to prove that f is "almost" additive. Given a small ϵ , we claim

$$f(z + \epsilon) + f(w) = f(z + \epsilon + w) \quad \text{for all } w > z \geq 133$$

By the condition, it's true for $w \in (z, z + z^2)$. Suppose it's true for $w \in (z, z + \frac{n}{2}z^2)$

Consider $w \in (z + \frac{n}{2}z^2 - \epsilon, z + \frac{n+1}{2}z^2)$. Note that

$$\sqrt{z + \frac{n+1}{2}z^2 + \epsilon + w} < \sqrt{z + \frac{n+1}{2}z^2 + \epsilon + z + \frac{n+1}{2}z^2} < 2z + nz^2$$

Thus,

$$f\left(\sqrt{z + \frac{n+1}{2}z^2 + \epsilon}\right) + f(w) = 2f(a)$$

where $a \in (z, z + \frac{n}{2}z^2)$. According to the inductive hypothesis,

$$f(z + \epsilon) + 2f(a) = f\left(\sqrt{z + \frac{n+1}{2}z^2 + \epsilon}\right) + 2f\left(\frac{z + \epsilon + w}{2}\right)$$

As the result, $f(z + \epsilon) + f(w) = 2f\left(\frac{z + \epsilon + w}{2}\right) = f(z + \epsilon + w)$, as desired.

We have already proved that $f(x) + f(y) = f(x + y)$ for large $x, y \in \mathbb{R}^+$. Thus, $f(x) = cx$ for all large $x \in \mathbb{R}^+$. Finally, because $f(2x) = 2f(x)$, $\forall x \in \mathbb{R}^+$, we get $f(x) = cx$, $\forall x \in \mathbb{R}^+$

In fact, in these examples, we just extend the "size" of a particular set. Formally, let

$$S_0 = \{(x, y) \mid f(x + y) = f(x) + f(y) \text{ is proved}\}$$

Our goal is to find sets $S_1, S_2, \dots, S_k$ such that

$$S_0 \subset S_1 \subset S_2 \subset \dots \subset S_k = \mathbb{R} \times \mathbb{R}$$

and it's easy to extend S_{i-1} to S_i . To illustrate this idea, we'll solve two problems.

Example 30. (APMO) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$(z + 1)f(x + y) = f(xf(z) + y) + f(yf(z) + x)$$

holds for all $x, y \in \mathbb{R}$

Example 31. (2014 Iran) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ st for all positive reals x, y ,

$$f\left(\frac{y}{f(x+1)}\right) + f\left(\frac{x+1}{xf(y)}\right) = f(y)$$

2.4 Applications

2.5 Problems

Problem 21. (2014 Iran) Find all continuous functions $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$f(xf(y)) + f(f(y)) = f(x)f(y) + 2, \forall x, y \in \mathbb{R}_{\geq 0}$$

Problem 22. (2002 A4) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$(f(x) + f(z))(f(y) + f(t)) = f(xy - zt) + f(xt + yz)$$

for all real x, y, z, t

Problem 23. (2003 A5) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(xyz) + f(x) + f(y) + f(z) = f(\sqrt{xy})f(\sqrt{yz})f(\sqrt{zx})$$

and $f(x) < f(y)$ when $1 \leq x < y$

Problem 24. (2009 A7) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(x+y)) = f(yf(x)) + x^2, \forall x, y \in \mathbb{R}$$

Problem 25. (2012 A5) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(1+xy) - f(x+y) = f(x)f(y), \forall x, y \in \mathbb{R}$$

and $f(-1) \neq 0$.

Problem 26. (2017 Iran) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$\frac{x + f(y)}{xf(y)} = f\left(\frac{1}{y} + f\left(\frac{1}{x}\right)\right)$$

Chapter 3

Double Counting

3.1 Switching

Basically, *LHS* and *RHS*, in the equation, must hold the same properties. For instance, if *LHS* is an increasing function of x , then *RHS* is also an increasing function of x . In particular, if *LHS* is symmetric w.r.t the variables, then so is *RHS*. If this is the case, we simply exchange the variables and observe what would happened.

Example 32. (Socrates) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + f(y)) = f(x^2) + 2x^2f(y) + f(y)^2$$

holds for all $x, y \in \mathbb{R}$

Solution. Notice that the left hand side is already symmetric. Switch x, y , we see that

$$f(x^2) + 2x^2f(y) + f(y)^2 = f(y^2) + 2y^2f(x) + f(x)^2$$

We now want to choose y that minimizes the number of uncertain values. After solving

$$y^2 - y = 0 \Leftrightarrow y = 0, 1$$

we tend to "take" two equations:

$$P(x, 1) \rightarrow f(x^2) + 2x^2f(1) + f(1)^2 = f(1) + 2f(x) + f(x)^2$$

$$P(x, 0) \rightarrow f(x^2) + 2x^2f(0) + f(0)^2 = f(0) + f(x)^2$$

From here, one can easily get the answers

$$f(x) = 0, \forall x \in \mathbb{R} \quad \text{or} \quad f(x) = x^2, \forall x \in \mathbb{R}$$

Indeed, switching the variables is equivalent to comparing $P(x, y), P(y, x)$. We usually get a simple identity, which is much easier to solve the function. The only bad news is: In most of the problems, *LHS* and *RHS* aren't symmetric.

To deal with this situation, we can assert a special form to one of the variables.

Example 33. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ so that

$$f(f(x) + y) = x + f(y), \forall x, y \in \mathbb{N}$$

Solution. As the previous comment, to make *LHS* become symmetric, substitute $f(y)$ for y , then we get

$$f(f(x) + f(y)) = x + f(f(y))$$

Now, the *LHS* is symmetric and we can switch x and y (Hooray!)

$$x + f(f(y)) = f(f(x) + f(y)) = f(f(y) + f(x)) = y + f(f(x))$$

This means

$$f(f(z)) = z + c, \forall z \in \mathbb{N}$$

for some constant c (why) Recall, in chapter one, I have mentioned: Always use the little result to get another identity. In this example,

$$P(x, f(y)) \rightarrow f(x + y + c) = f(x) + f(y)$$

So f is eventually linear. Like what we do in chapter three: for any given x , take a sufficient large y , we can see that $x, f(x)$ satisfy the same linear equation and we conclude that f is linear. The rest is easy.

Remark 6. In chapter one, we solve this FE by using the injectivity.

Example 34. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the following equation

$$f(xf(y)) - x = f(xy), \forall x, y \in \mathbb{R}$$

Solution. It seems that we could do our magic trick

$$f(xf(y)) - x = f(xy) = f(yx) = f(yf(x)) - y$$

This identity isn't very useful. Thus, we need to find another one. Rewrite the condition

$$f(xf(y)) = f(xy) + x$$

Replace x by $f(x)$,

$$f(f(x)f(y)) = f(f(x)y) + x = f(xy) + y + f(x)$$

Since now LHS is symmetric,

$$f(xy) + y + f(x) = f(yx) + x + f(y)$$

must hold. And thus, f must be linear (why) Assume $f(x) = ax + b$, we have

$$\begin{aligned} LHS &= axf(y) + b = ax(ay + b) + b = a^2xy + abx + b \\ RHS &= axy + b + x \end{aligned}$$

Compare the coefficient, $a = 1, b = 1$. In other words,

$$f(x) = x + 1, \forall x \in \mathbb{R}$$

Example 35. (2016 Iran) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for all $a, b \in \mathbb{N}$, we have

$$(f(a) + b)f(a + f(b)) = (a + f(b))^2$$

Solution. Similar to example 33, we have

$$(f(f(a)) + b)f(f(a) + f(b)) = (f(a) + f(b))^2$$

As the result,

$$f(f(n)) = n + c$$

for some constant c . Note that f has fixed point:

$$P(a, a) \rightarrow f(a + f(a)) = a + f(a)$$

It follows that $f(f(n)) = n$ for all $n \in \mathbb{N}$ Now,

$$(f(a) + f(b))f(a + b) = (a + b)^2$$

The rest is easy. [Example 6](#)

Although this trick is useful, there are still many problems can't be solved in this way. For example,

Example 36. (Based on: 2014 Romania) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x + 3f(y)) = f(x) + f(y) + 2y, \forall x, y \in \mathbb{R}^+$$

Fortunately, if we modify our trick a little bit, the previous FE becomes trivial.

3.2 Three variables method

As the previous discussion, we hope to find a more useful trick. Our idea is simple! Usually, we got stuck after switching variables because there are many restrictions. Consequently, we introduce a "free" variable in order to ignore the restrictions.

Example 37. (Socrates) Find all solutions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ satisfying the equation

$$f(x+y)^2 = f(x)^2 + 2f(xy) + f(y)^2, \forall x, y \in \mathbb{R}^+$$

Solution. Despite the consequences, replace y by $y+z$:

$$f(x+y+z)^2 = f(x)^2 + 2f(x(y+z)) + f(y+z)^2$$

Notice that the term $f(y+z)^2$ can be further simplified:

$$f(y+z)^2 = f(y)^2 + 2f(yz) + f(z)^2$$

As the result,

$$f(x+y+z)^2 = f(x)^2 + f(y)^2 + f(z)^2 + 2f(x(y+z)) + 2f(yz)$$

Cyclic the variables, we get

$$f(yz) + f(xy+zx) = f(zx) + f(yz+xy) = f(xy) + f(zx+yz)$$

Since there is always a solution to the system of equations: $yz = a, zx = b, xy = c$, the identity can be rewrote as

$$f(a) + f(b+c) = f(b) + f(c+a) = f(c) + f(a+b)$$

Fix b, c , we see that

$$f(a+c) - f(a) = f(b+c) - f(b)$$

In other words,

$$f(x+c) = f(x) + (f(b+c) - f(b))$$

Define $g : \mathbb{R}^+ \rightarrow \mathbb{R}$ by $g(c) = f(b+c) - f(b)$, then

$$f(x+y) = x + g(y), \forall x, y \in \mathbb{R}^+$$

which is an easy exercise.

Remark 7. Note that the function g is well-defined

Example 38. (Based on: 2014 Romania) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x + 3f(y)) = f(x) + f(y) + 2y, \forall x, y \in \mathbb{R}^+$$

Solution. Observe the equation

$$f(x + 3f(y + z)) = f(x) + f(y + z) + 2(y + z)$$

We hope that there is another way to calculate LHS. According to the condition, we tend to replace z by $3f(z)$:

$$\begin{aligned} LHS &= f(x + 3f(y) + 3f(z) + 6z) \\ &= f(x + 3f(y) + 6z) + f(z) + 2z \\ &= f(x + 6z) + f(y) + 2y + f(z) + 2z \end{aligned}$$

On the other hand,

$$\begin{aligned} RHS &= f(x) + f(y + 3f(z)) + 2y + 6f(z) \\ &= f(x) + f(y) + f(z) + 2z + 2y + 6f(z) \end{aligned}$$

We conclude that

$$f(x + 6z) = f(x) + 6f(z)$$

holds for all $x, y \in \mathbb{R}^+$. Now,

$$f(6x + 6z) = f(6x) + 6f(z) \rightarrow 6f(z) = f(6z) + c, \forall z \in \mathbb{R}^+$$

for some constant c . Therefore,

$$f(x + z) = f(x) + f(z) + c$$

which implies f is linear (why) and thus,

$$f(x) = x, \forall x \in \mathbb{R}^+$$

Remark 8. Because we want to calculate an expression in two different ways, we usually "iterate" our condition. Besides, since we exchange two variables, we usually need to deal with something like Cauchy FE and get the additivity.

Example 39. (Socrates) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(2x + 2f(y)) = x + f(x) + 2y$$

holds for all $x, y \in \mathbb{R}^+$

Solution. Substitute $2y + 2f(z)$ for y ,

$$\begin{aligned} LHS &= f(2x + 2f(2y + 2f(z))) \\ &= f(2x + 2y + 2f(y) + 4z) \\ &= x + y + 2z + f(x + y + 2z) + 2y \end{aligned}$$

On the other hand,

$$\begin{aligned} RHS &= f(2x + 2f(2y + 2f(z))) \\ &= x + f(x) + 2(2y + 2f(z)) \end{aligned}$$

After simplification,

$$f(x + y + 2z) + 2z = f(x) + y + 2f(z)$$

By switching the variables, we see that f is linear and thus,

$$f(x) = x, \forall x \in \mathbb{R}^+$$

Remark 9. In fact, we can solve this problem by switching x, y directly. We have

$$f(2f(x) + 2f(y)) = f(x) + f(f(x)) + 2y$$

Thus,

$$f(x) + f(f(x)) = 2x + c$$

for some constant c . According to [Example 20](#), we conclude that

$$f(x) = x, \forall x \in \mathbb{R}^+$$

However, switching doesn't help for the following problem:

Determine all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ so that

$$f(2x + 2f(y)) = x + f(x) + 2y$$

holds for all positive reals x, y

In fact, I'm inspired by

Example 40. (2007 A4) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ so that

$$f(x + f(y)) = f(x + y) + f(y), \forall x, y \in \mathbb{R}^+$$

Solution. The solution is due to Andreas Dwi Maryanto Gunawan, from Indonesia.
For any positive reals z , we have that

$$f(x + f(y)) + z = f(x + y) + f(y) + z$$

Then, take f on both side

$$f(f(x + f(y)) + z) = f(f(x + y) + f(y) + z)$$

Use the condition to reduce expressions:

$$f(x + f(y) + z) + f(x + f(y)) = f(x + y + f(y) + z) + f(x + y)$$

Again, we have

$$\begin{aligned} f(x + y + z) + f(y) + f(x + y) + f(y) &= f(x + 2y + z) + f(y) + f(x + y) \\ \Rightarrow f(x + y + z) + f(y) &= f(x + 2y + z) \end{aligned}$$

The rest is a routine work.

Remark 10. In general, if our condition is

$$f(x + g(y)) = g(y) + h(x, y)$$

for some functions g, h . Then we can try to compare $P(x, y + g(z))$ and $P(x + h(y, z), y)$

Exercise. (Socrates) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ so that for all $x, y \in \mathbb{R}^+$

$$f(x + f(y)) = 3f(x) - 2x + f(y)$$

Sometimes, we need to use this method twice.

Exercise. (2016 Taiwan) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x + y + f(y)) = 4030x - f(x) + f(2016y), \forall x, y \in \mathbb{R}^+$$

3.3 Double Counting

Indeed, switching and three variables method are kinds of double counting. The central concept is: for a particular expression, we calculate the value in two different ways.

Example 41. (2015 APMO) Let $S = \{2, 3, 4, \dots\}$ denote the set of integers that are greater than or equal to 2. Does there exist a function $f : S \rightarrow S$ such that

$$f(a)f(b) = f(a^2b^2) \quad \text{for all } a, b \in S \text{ with } a \neq b$$

Solution. Notice that we have

$$\begin{aligned} f(2^1)f(2^2)f(2^3)f(2^4)f(2^5) &= f(2^3)f(2^4)f(2^5)f(2^6) \\ &= f(2^5)f(2^6)f(2^{14}) \\ &= f(2^{14})f(2^{22}) \\ &= f(2^{72}) \end{aligned}$$

On the other hand,

$$\begin{aligned} f(2^1)f(2^2)f(2^3)f(2^4)f(2^5) &= f(2^1)f(2^2)f(2^4)f(2^{16}) \\ &= f(2^2)f(2^4)f(2^{34}) \\ &= f(2^4)f(2^{72}) \end{aligned}$$

So $f(2^4) = 1$, which is absurd! Thus, we conclude that there is not such a function.

Remark 11. If we apply the following transformation: Let $g : \log S \rightarrow \log S$ defined by

$$g(x) = \log f(e^x)$$

Then we find that the condition becomes

$$g(a) + g(b) = g(2a + 2b)$$

Consequently,

$$g(a) + g(b) + g(2c) = g(2a + 2b) + g(2c) = g(4a + 4b + 4c)$$

Switching the variables, we get

$$g(2d) = g(d) + c, \quad \forall d \in \log S$$

And so,

$$2g(d) = g(4d) = g(2d) + c = g(d) + 2c \rightarrow g(d) = 2c$$

It follows that $g \equiv 0$, which is impossible!

Of course, we need to have some identities to "open" the terms. Notice that we, initially, have only the condition. Therefore, we usually use the given condition to do the trick.

Example 42. (2014 Poland) Find all functions $f : \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ satisfying

$$\underbrace{f(f(f(\dots f(f(q)) \dots)))}_n = f(nq)$$

for all integers $n \geq 1$ and rational numbers $q > 0$

Solution. We want to calculate the following expression

$$\underbrace{f(f(f(\dots f(f(q)) \dots)))}_n$$

in another way. By the definition of composition,

$$LHS = \underbrace{f(f(f(\dots f(f(f(f(f(\dots f(f(q)) \dots)))))) \dots))}_a \underbrace{}_{n-a}$$

It follows that

$$f(a(n+1-a)q) = f(nq)$$

holds for all $a \in \{1, 2, \dots, n\}$. In particular,

$$f(a(a^2 - a + 1 - a)q) = f((a^2 - a)q) \rightarrow f(q) = f(mq), \forall m \in \mathbb{N}$$

by simple substitution. Now, for any $x, y \in \mathbb{Q}^+$, write x, y as $\frac{q}{r}, \frac{q'}{r'}$, respectively. Then

$$f(x) = f(q'r \cdot x) = f(qq') = f(qr' \cdot y) = f(y)$$

We conclude that f is constant, as desired.

Remark 12. Since we expect the only function is constant, we may want to show that for any $x, y \in \mathbb{Q}^+$, there are some $a < n \in \mathbb{N}$ and $q \in \mathbb{Q}^+$ such that

$$\begin{cases} x = a(n-a)q \\ y = nq \end{cases}$$

This is not so obvious. The readers can try to give it a proof or a counter example.

In most situations, it suffices to find out "short" (because we hope to get a simple result) identities in the beginning.

Example 43. (Socrates) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ so that

$$f(x^3) - f(y^3) = (x - y)(f(x^2) + f(xy) + f(y^2)), \forall x, y \in \mathbb{R}^+$$

Solution. WLOG, assume $f(1) = 1$. Notice that

$$f(x^3) - f(z^3) = (f(x^3) - f(y^3)) + (f(y^3) - f(z^3))$$

Therefore, we have

$$\begin{aligned} (x^2 - 1)(f(x^4) + f(x^2) + 1) &= (x^2 - x)(f(x^4) + f(x^3) + f(x^2)) \\ &\quad + (x - 1)(f(x^2) + f(x) + 1) \end{aligned}$$

After simplification,

$$f(x^4) = x(x - 1)f(x^2) + (x^2 - x + 1)f(x) + x(x - 1)$$

For convenience, let g, h be two polynomials such that

$$f(x^3) = g(x, f(x), f(x^2)) \quad \text{and} \quad f(x^4) = h(x, f(x), f(x^2))$$

Our idea is to calculate $f(x^{12})$ in two ways. Since $(x^4)^3 = x^{12} = (x^3)^4$, it follows

$$\begin{aligned} f(x^{12}) &= h(x^3, f(x^3), f(x^6)) \\ &= h(x^3, g(x, f(x), f(x^2)), g(x^2, f(x^2), f(x^4))) \\ &= h(x^3, g(x, f(x), f(x^2)), g(x^2, f(x^2), h(x, f(x), f(x^2)))) \end{aligned}$$

and

$$\begin{aligned} f(x^{12}) &= g(x^4, f(x^4), f(x^8)) \\ &= g(x^4, h(x, f(x), f(x^2)), h(x^2, f(x^2), f(x^4))) \\ &= g(x^3, h(x, f(x), f(x^2)), h(x^2, f(x^2), h(x, f(x), f(x^2)))) \end{aligned}$$

From here, one can show that

$$f(x^2) + f(x) = x^2 + x, \quad \forall x \in \mathbb{R}^+$$

Finally,

$$f(x^3) = (x - 1)(f(x^2) + f(x) + 1) + 1 = (x - 1)(x^2 + x + 1) + 1 = x^3$$

Usually, we use two operations which are commute to realize our idea.

3.4 Exercise

We begin with standard exercises.

Problem 27. (2006 MOP) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(x + f(y)) = x + f(f(y))$$

for all real numbers x and y

Problem 28. Determine all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x + y + f(y)) + f(x + z + f(z)) = f(2f(z)) + f(2y) + f(2x)$$

holds for all $x, y, z \in \mathbb{R}^+$

Problem 29. (Socrates) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x + f(x) + y) = f(y) + 2x, \quad \forall x, y \in \mathbb{R}^+$$

Problem 30. (Socrates) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x + 2f(y)) = f(x + y) + y, \quad \forall x, y \in \mathbb{R}^+$$

Problem 31. (Mohammed Jafari) Determine all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x + f(x) + 2y) = f(2x) + y + f(y)$$

holds for all $x, y \in \mathbb{R}^+$

Problem 32. (2007 Romania) Let $f : \mathbb{Q} \rightarrow \mathbb{R}$ be a function such that

$$|f(x) - f(y)| \leq (x - y)^2$$

for all $x, y \in \mathbb{Q}$. Prove that f is constant.

Problem 33. (Own) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfying that for all $a, b \in \mathbb{N}$,

$$f(a + f(b)) = f(a)^{f(b)}$$

Problem 34. (MDS) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x + f(y)) + x = f(x + f(x)) + y$$

holds for all positive reals x, y

In fact, for most problems, double counting is used to find some useful identities. Try to bash the following problems! :)

Problem 35. (Crux) Find all real functions f such that

$$f(xy) = f(f(x) + f(y))$$

holds for all real x, y

Problem 36. (2017 Iran) Find all functions $f : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$ that satisfy the following conditions: for all positive reals x, y, z ,

$$f(f(x, y), z) = x^2 y^2 f(x, z)$$

and

$$f(x, 1 + f(x, y)) \geq x^2 + xyf(x, x)$$

Problem 37. (2010 A5) Determine all functions $f : \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ which satisfy the following equation for all $x, y \in \mathbb{Q}^+$:

$$f(f(x)^2 y) = x^3 f(xy)$$

Problem 38. (Socrates) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x^2 + yf(z) + f(x)) = xf(x) + zf(y) + x, \quad \forall x, y, z \in \mathbb{R}^+$$

Problem 39. (2005 A2) Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ which have the property:

$$f(x)f(y) = 2f(x + yf(x))$$

for all positive real numbers x and y

Problem 40. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x)f(yf(x) - 1) = x^2 f(y) - f(x), \quad \forall x, y \in \mathbb{R}$$

Problem 41. (2013 USA) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ that satisfy the equation

$$f^{abc-a}(abc) + f^{abc-b}(abc) + f^{abc-c}(abc) = a + b + c$$

for all $a, b, c \geq 2$